

1 Life-history trade-offs in the regeneration niche: Post-fire 2 seeding vs. resprouting in Fynbos *Proteaceae*

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6 Abstract

7 Plant regeneration in fire-driven ecosystems lies in a continuum between
8 obligate reseeders, whose recruits must grow rapidly to reproduce before
9 the next fire, and resprouters, which invest in fire-tolerance traits. Previous
10 population persistence models explained coexistence via
11 reproduction/resilience trade-offs in fire-prone ecosystems but used
12 carbon allocation assumptions with limited empirical support. We
13 combined meteorological and leaf-level gas exchange measurements in a
14 fire-prone summer-drought ecosystem to quantify key ecophysiological
15 traits in a reseeder and a resprouter Fynbos *Protea* species three years
16 postfire. Stomatal conductance (g_{sw}) was the primary trait distinguishing
17 types: reseeders initiated photosynthetic activity earlier in spring and had
18 g_{sw} responsive to vapour pressure deficit, light, soil moisture, and CO₂,
19 enabling faster resource acquisition during cool and wet periods relative to
20 resprouter recruits, but shifting into low photosynthetic activity or
21 respiration when soils dried. These findings provide a mechanistic basis for
22 the reproduction/resilience hypothesis and challenge prevailing
23 assumptions by demonstrating an isohydric strategy in reseeders,
24 suggesting a previously unrecognized drought-avoidance in these species,
25 while also indicating an interaction between drought and disturbance-
26 related life history traits. These divergent strategies facilitate an ecosystem-
27 scale transition from reseed-dominated photosynthesis during the cooler
28 wet season to resprouter-driven productivity during summer drought.

29 **Keywords:** *reseeding, resprouting, Mediterranean climate, ecophysiology,*
30 *fire recovery, fire, drought*

31 Introduction

32 Understanding functional trade-offs in plant life history is of fundamental
33 importance in attempts to define plant functional types for implementing
34 models of vegetation and ecosystem responses to changing climate,
35 disturbance and resource regimes. It is well known that current dynamic
36 global vegetation model (DGVMs), which are widely used tools in global
37 carbon budget modelling, are limited in their ability to represent the carbon-
38 dependent growth responses of disturbance driven ecosystems (Argles et al.
39 2022; O'Sullivan et al. 2024), and their resilience to large scale disturbance
40 regimes like wildfire (Magadzire et al. 2019). This resilience is strongly
41 determined by plant species' life history characteristics in the regeneration
42 niche, the critical period post-disturbance that defines the development
43 trajectory of plant communities and entire ecosystems between disturbance
44 events (Johnstone et al. 2016).

45 Because episodic wildfire disturbance is a dominant feature of about 40% of
46 the terrestrial land surface, DGVM tools have pursued efforts to take such
47 behaviours into account, but are limited by a lack of primary data to
48 implement the necessary algorithms (Scheiter et al. 2013). More credible
49 simulation of divergent plant life history behaviours in the regeneration
50 niche remains unresolved, even though this may strongly determine the rate
51 of recovery of ecosystem carbon gain and carbon stocks post fire. This is an
52 important gap, because the timing and duration of fire seasons are changing
53 globally as a result of climate and global change, with significant
54 implications for divergent plant functional types (Tangney et al. 2022).

55 Two primary fire adaptation strategies exist: reseeding and resprouting
56 (Treurnicht et al. 2020; Bond and Midgley 2001; Altwegg et al. 2015).
57 Reseeders tend to prioritize rapid growth and reproduction, showing higher

58 photosynthetic rates, which allows them to exploit the nutrient-rich post-
59 fire environment and replenish seed banks before subsequent fires (Rogers
60 et al. 2024; Parra and Moreno 2017; Hernández et al. 2011; Vilagrosa et al.
61 2014). These species often maintain higher stomatal conductance, even in
62 the afternoon (Clemente et al. 2005; van Blerk et al. 2021; van Wilgen et al.
63 1992), enhancing CO₂ uptake for faster growth despite increased water loss
64 (Parra and Moreno 2017; van Wilgen et al. 1992), following a ‘high risk -
65 high reward’ strategy. While often associated with high gas exchange, some
66 evidence suggests that under extreme water stress, seedlings may
67 temporarily exhibit isohydric-like behaviour, regulating stomata
68 conservatively to maintain leaf water potential and avoid hydraulic failure
69 (Roman et al. 2015; Manzoni et al. 2013). For instance, *Cistus ladanifer*
70 seedlings in the first summer post-fire displayed greater predawn water
71 potential, leaf conductance, and photosynthesis compared to mature plants
72 (Clemente et al. 2005).

73 Resprouters, on the other hand, are thought to mobilize stored reserves for
74 resprouting, favouring adult-stage persistence and greater resource
75 conservation (Parra and Moreno 2017; Altwegg et al. 2015; Lawes et al.
76 2022) relative to rapid growth (van Blerk et al. 2021). Due to their
77 established root systems, they are often capable of sustaining gas exchange
78 longer during drought, letting stomatal conductance decrease gradually
79 following an anisohydric strategy, where plants tolerate lower water
80 potentials to maintain carbon assimilation (Roman et al. 2015; Manzoni et
81 al. 2013). This is assumed to be due to access to deeper water sources via an
82 established root system (Parra and Moreno 2017; van Blerk et al. 2021;
83 Clemente et al. 2005). Indeed, resprouters allocate nearly twice as much
84 biomass to roots as seedlings and recruits in the regeneration stage (Poorter
85 et al. 2012). This is likely a trade-off that reduces above-ground growth and
86 investment in leaf area, thus reducing whole plant growth rate and biomass
87 accumulation (Altwegg et al. 2015). However, emerging evidence suggests

88 that these hydraulic strategies may not align strictly with regeneration
89 mode, highlighting the need for species-specific evaluation.

90 More conservative regeneration strategies in resprouters appear to be
91 accompanied by lower photosynthetic rates compared to reseeders
92 (Hernández et al. 2011; Rogers et al. 2024; van Wilgen et al. 1992). However,
93 the evidence is mixed, with some resprouter seedlings shown to exhibit a
94 higher ETR compared to reseeders seedlings (Hernández et al. 2011). In
95 contrast to the conservative recruits of resprouter species, resprouting
96 adult plants are reported to increase photosynthetic rate due to enhanced
97 nutrient availability and reduced stomatal limitations (Rogers et al. 2024).
98 For instance, some *Eucalyptus* species increase their stomatal conductance
99 post-fire, leading to up to a 20% increase in photosynthesis compared to
100 unburnt trees (Rogers et al. 2024).

101 Shifts in fire regime, driven largely by climate and land use changes, may
102 extend fire seasons beyond historical limits, driven by earlier drought onset
103 (Tangney et al. 2022). Fires occurring during sensitive phenological stages
104 (such as growth, dormancy, flowering, or reproduction) can also hinder
105 post-fire recovery (Miller et al. 2019). Hence, an increase in post-fire
106 survival in resprouting species has previously been observed after fires
107 outside of the historical fire season (Tangney et al. 2022), with a preference
108 for medium-length fire intervals (Nolan et al. 2021), e.g. 5-24 years in
109 Australian shrublands (Enright et al. 2014). The capacity of plant species or
110 communities to adapt depends on the pace of these changes (Tangney et al.
111 2022). In the fire-prone species-rich Fynbos, the interplay of shifting fire
112 regime and changing drought risk appear to be an important explanation for
113 recent shifts in plant functional type dominance towards resprouting
114 (Slingsby et al. 2017).

115 Shifting fire frequency, intensity and seasonality interact strongly with
116 divergent plant regeneration strategies. Resprouting species are especially
117 vulnerable to increasing fire intensity, where hotter and larger fires than the

118 norm can increase mortality of plants that typically survive fire (Simpson et
119 al. 2021; Kruger and Bigalke 1984; van Blerk et al. 2021). Conversely,
120 reseeding species, which depend on post-fire recruitment, are more
121 susceptible to rapid increases in fire frequency due to their dependence on
122 long-lived seed banks and the time required to reach mature reproductive
123 stages (Tangney et al. 2022; Altwegg et al. 2015). As a result, the
124 physiological adaptations of both types differ somewhat.

125 For instance, seasonal variations in leaf water content were usually higher
126 in reseeders than in resprouters and were associated with higher resistance
127 to embolism, while resprouters could sustain a positive carbon balance
128 during drought periods (Nefzi et al. 2023). Note that these adaptations often
129 also exist in species that persist in areas frequently exposed to drought
130 (McDowell et al. 2008), although the specific adaptations of physiological
131 parameters may be different to the ones required for recovery after fire.

132 Similarly, community assembly in fire-prone ecosystems is controlled on the
133 one hand by differing resource use (Tilman 1982) through trait divergence
134 that limits similarity (Weiher and Keddy 2001; Nottebrock et al. 2017), and
135 by the environmental filtering of the fire regime: Under a lower fire
136 frequency, reseeding species tend to dominate (Altwegg et al. 2015; Lawes
137 et al. 2022; Simpson et al. 2021; Bond and Midgley 2001), as it takes them
138 more time to mature and produce seed (Altwegg et al. 2015; Kruger and
139 Bigalke 1984). Hence, reseeders are typically thought to be selected by an
140 intermediate fire frequency (Altwegg et al. 2015). Note that reseeding
141 grasses may not be as disadvantaged as reseeding woody plants due to their
142 lack of investment in carbon-dense stems and branches, and thus appear to
143 prefer higher fire frequency that may disadvantage long-lived resprouting
144 grasses (Simpson et al. 2021).

145 Numerous ecological studies on community structure and regeneration
146 after fire exist (Tangney et al. 2022; Miller et al. 2019; Menges et al. 2020).
147 However, previous studies have either focused on leaf structural traits or

148 evolutionary divergence to explain the prevalence of reseeders or
149 resprouters in fire-prone ecosystems, such as was done in the *Proteaceae*
150 family to infer ecological diversity (Nolting et al. 2021; Onstein et al. 2016).
151 Similarly, Altwegg et al. (2015) used a demographic modelling approach
152 with an assumed functional trade-off to simulate how resprouters may be
153 more common than reseeders in regions with short fire return intervals.

154 Empirical studies linking plant physiological parameters to the divergent
155 strategies of resprouters and reseeders remain scarce. Here, we examine the
156 ecophysiological parameters that govern leaf function in two dominant, co-
157 occurring species from the same genus (*Protea*) with differing regeneration
158 strategies, which minimizes phylogenetic confounding and helps isolate
159 traits specific to each. Following previous findings (Altwegg et al. 2015), we
160 hypothesize that a trade-off exists between the two strategies (the
161 reproduction/resilience trade-off hypothesis): (1) reseeders maximize
162 carbon gain during favorable pulses through higher stomatal conductance
163 (g_{sw}) that is highly sensitive to environmental drivers, and (2) resprouters
164 maintain physiological resilience during drought via hydraulic decoupling
165 from surface moisture, characterized by g_{sw} that is less sensitive to vapor
166 pressure deficit (VPD) and soil water content (SWC).

167 Materials & Methods

168 *Site description & above-canopy meteorological measurements*

169 The Jonkershoek Fynbos research site (33°59'25.0"S, 18°57'19.6"E) is
170 located near the city of Stellenbosch, South Africa, in the Swartboskloof
171 valley (Supplementary Figure S1), forming part of the Jonkershoek Nature
172 Reserve. The reserve contains a variety of Fynbos ecosystems, which are fire
173 driven with a fire return period of *ca.* 10-13 years (van Wilgen 2009), with
174 the last wildfire dating to February 2021 (CapeNature 2021). Fynbos
175 species are adapted to this fire regime, either through serotinous cones from
176 which seeds are released after fire, with seeds that germinate after heat or

177 smoke exposure, or through plant organs that can resprout (van Blerk et al.
178 2021). Raw eddy covariance (EC) flux data was obtained from the research
179 site at Swartboskloof – operated by the South African ‘Expanded Freshwater
180 and Terrestrial Environmental Observation Network’ (EFTEON;
181 <https://efteon.saeon.ac.za/>) – which provided auxiliary meteorological
182 parameters for this study (radiation, CO₂ concentration, soil water content
183 at 10cm depth extracted to match timing of leaf level measurements).

184 Swartboskloof is a fan-shaped valley with a north-facing aspect, ranging
185 from 285 to 1200m above sea level (van Wilgen et al. 1992). The mean
186 annual temperature and precipitation in Swartboskloof was $17.2 \pm 0.3^{\circ}\text{C}$ and
187 $1858 \pm 483\text{mm}$ for the years 2019-2023 at the weather station situated at the
188 research site, respectively (excluding 2021 due to missing data). Both values
189 show a marked increase from measurements from the 1980s showing
190 16.2°C and 1523mm , respectively (van Wilgen et al. 1992). This
191 Mediterranean-type climate is characterised by hot dry summers from
192 November to March (Mean temperature in January: $22.9 \pm 1.8^{\circ}\text{C}$, increased
193 by 1.8°C from the 1980s at *ca.* 21.1°C in February; van Wilgen and Forsyth
194 1992). The observed temperature increase indicates a shift from a warm-
195 summer to a hot-summer Mediterranean-type climate (Köppen climate
196 class Csa; Peel et al. 2007). Mean monthly precipitation was 28-58mm in
197 summer (Dec-Feb) and 284-374mm in winter (Jun-Aug), respectively, with
198 monthly winter temperatures averaging $12.6\text{-}14.1^{\circ}\text{C}$. The mean monthly
199 maximum vapour pressure deficit was at its highest of $4.7 \pm 0.7\text{ kPa}$ in
200 February, and at its lowest of $2.4 \pm 0.3\text{ kPa}$ in July, for the 2019-2023 period.
201 Fynbos is characterised by nutrient depauperate, quartzite and sandstone
202 soils (Rebelo et al. 2006). These sandy water-repellent soils lead to plant
203 water stress, especially in summer, in spite of the high winter precipitation,
204 the (van Wilgen et al. 1992).

205 *Leaf-level measurements*

206 Measurements were carried out between May 2023 and January 2024
207 period, thus covering all main seasons. We selected two co-occurring,
208 dominant species from the same genus (*Protea*) to represent the two
209 strategies. By using closely related taxa (Resprouter: *Protea nitida* Mill., and
210 reseeders: *Protea repens* L.), we minimized phylogenetic confounding,
211 allowing for a clearer isolation of the physiological traits specifically
212 associated with the reseedling and resprouting strategies. Seedlings and
213 adults were differentiated (*P. repens* adults were not available due to the
214 recent fire in 2021), with plants along the transect selected randomly. This
215 resulted in three plant types: *P. nitida* adult & seedlings, and *P. repens*
216 seedlings (labelled *Nit-A*, *Nit-S* and *Rep-S*, respectively).

217 Stomatal conductance (g_{sw}), chlorophyll fluorescence measurements,
218 providing electron transfer rate (ETR) and photosystem II efficiency (Φ_{PSII}),
219 were performed every two weeks on all plant types using a portable leaf
220 porometer/fluorometer (LI-600F, LI-COR Biosciences, Lincoln, NE, USA).
221 Five leaves on 5 individuals were sampled with the LI-600F for each type in
222 the youngest 3 consecutive yearly leaf age groups, termed 'young', 'mid' and
223 'old' (*Yng*, *Mid*, *Old*), where old leaves were the first to emerge after the last
224 wildfire. Sampling was done in the morning (between 9:00-10:00), noon
225 (12:00-13:00), and afternoon (15:00-16:00).

226 A complementary data set comprising light-response curves (LRC) was
227 measured once in winter, spring and summer, using a portable
228 photosynthesis measurement system (LI-6400, LI-COR Biosciences, Lincoln,
229 NE, USA). One light response curve per type (*Nit-A*, *Nit-S* & *Rep-S*) and leaf
230 age (*Yng*, *Mid*, *Old*) was captured using the LI-6400 on 4 sampling dates (*Sep*:
231 2023-09-13, *Nov*: 2023-11-21, *Dec*: 2023-12-18 & *Jan*: 2024-01-25),
232 corresponding to early and late summer and spring, respectively,
233 representing 36 groups (3 types $\times$ 3 leaf ages $\times$ 4 measurement dates).
234 Photosynthetic parameters (A_{sat} and k_m) were derived by fitting a Marshall-

235 Biscoe model (photosynthesis R package; Stinziano et al. 2024), except for
236 plants exhibiting net respiration (specifically, Rep-S in December).
237 Additionally, leaf chlorophyll content of 3 leaves each in the 3 consecutive
238 yearly leaf age groups in each plant group was measured once using a
239 Decagon CCM300 (Opti-Sciences, Inc., Hudson NH, USA).

240 All measurements were carried out within 30-50m of the EC system located
241 at this site, in a transect extending in a wedge shape from the EC setup along
242 the dominant wind direction in winter (north-westerly transect; van Wilgen
243 et al. 1992) to link directly with meteorological data collected throughout
244 the majority of the sampling period. Despite potential summer footprint
245 mismatch of the transect caused by a shift to SE winds, high vegetation
246 uniformity ensured that physiological measurements remained
247 representative of the broader ecosystem.

248 *Data processing & statistical analyses*

249 EC fluxes were calculated using the EddyPro software (LI-COR Biosciences
250 2021). Datasets from the LI-600F and the LI-6400 were each merged with
251 EddyPro outputs and averaged in Python (Python Team 2023) using the
252 Pandas (McKinney 2010; Lenth 2025), Numpy (Harris et al. 2020), and Scipy
253 (Virtanen et al. 2020) libraries. All figures were created using ggplot2
254 (Wickham 2016) and ggpubr (Kassambara 2023) in R version 4.3.2 (R Core
255 Team 2023).

256 Significance of differences between plant types within each season was
257 analysed using generalized linear mixed-effects models fitted with
258 glmmTMB (Brooks et al. 2017; McGillicuddy et al. 2025), where g_{sw} and ETR
259 were modelled with a Gaussian distribution and Φ_{PSII} with a Beta
260 distribution. Models included species, season, and their interaction as fixed
261 effects, daytime as a covariate, and date as a random intercept, and pairwise
262 species contrasts within each season were obtained using emmeans (Lenth
263 2025) with Holm-adjusted p-values based on large-sample (Wald) tests.

264 Light response curves were fitted using the R package 'photosynthesis'
265 (Stinziano et al. 2021). Outliers were removed from LRC data to improve fit
266 (Fig. 1), with up to 5 points omitted to obtain a reasonable fit. As shown,
267 these points closely align with the fitted curves, indicating that the fitting
268 algorithm encountered difficulties in automatically generating an
269 appropriate curve (a known limitation of the 'photosynthesis' package).
270 Subsequently, the maximum (saturating) rate (A_{sat}) and light sensitivity of
271 photosynthesis (k_m ; i.e. how quickly the curve rises with increasing light)
272 were obtained by fitting a Marshall & Biscoe light-response function
273 (aq_response) to net photosynthesis (A_{net}) versus photosynthetically active
274 radiation (PAR) for each group (3 types $\times$ 3 leaf ages $\times$ 4 measurement
275 dates). A_{sat} was taken as $k_{sat} - R_d$ from the fitted parameters, and k_m was
276 computed numerically as the irradiance (PAR) at which net photosynthesis
277 equals half the light-saturated net rate ($0.5 \times A_{sat}$) using a root-finding
278 procedure (uniroot with an expanding upper bound; grid-
279 search/approximation used as a fallback). Group-specific slopes and
280 intercepts were estimated and compared using the emmeans R package
281 (Lenth 2025).

282 *Generalised Additive Model (GAM)*

283 A generalized additive model (GAM) was developed in R using the mgcv
284 package (Wood 2017). Model selection is described in detail in
285 Supplementary Methods S1. The selected model was used to examine the
286 non-linear response of stomatal conductance (g_{sw}) to environmental
287 parameters. The final model used the median data of the 5 sampled
288 individuals in each measurement (date, daytime, leaf age, plant group) and
289 included leaf vapour pressure deficit (VPD), soil water content (SWC), CO_2
290 mole fraction in the air (x_{CO_2}) and photosynthetically active radiation (PAR).

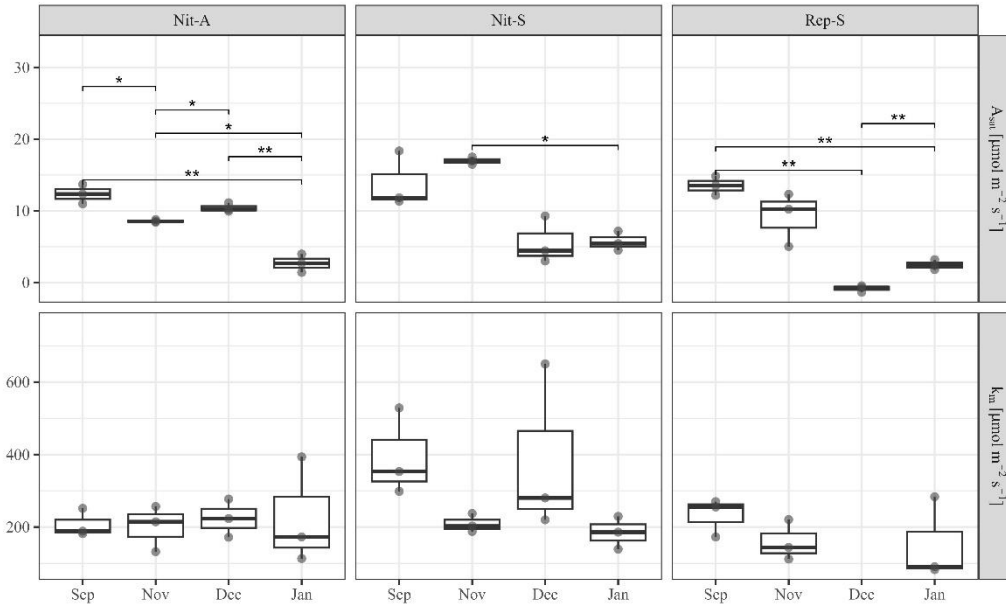
291 Results

292 **Seasonal Dynamics of Photosynthetic Capacity:** Light response curves
293 (LRCs) revealed distinct seasonal trajectories in photosynthetic capacity
294 (A_{sat}) and light sensitivity (sometimes called 'half-saturation light level', k_m).
295 Generally, A_{sat} peaked in spring, followed by a sharp summer decline across
296 all groups (Figure 1 & Supplementary Figure S2).

297 Seedlings of the reseeder (*P. repens*, Rep-S) exhibited the highest drought
298 sensitivity, with A_{sat} peaking in early spring (Sept: $13.53 \pm 1.33 \mu\text{mol m}^{-2} \text{s}^{-1}$,
299 Figure 1), before shifting into net respiration with the onset of summer
300 drought (Dec: $-0.68 \pm 0.47 \mu\text{mol m}^{-2} \text{s}^{-1}$). A minor recovery in January
301 ($2.38 \pm 0.70 \mu\text{mol m}^{-2} \text{s}^{-1}$) did not restore significant carbon assimilation
302 capacity. Conversely, resprouter seedlings (*P. nitida*, Nit-S) followed a
303 delayed trajectory, peaking in late spring (Nov: $16.95 \pm 0.50 \mu\text{mol m}^{-2} \text{s}^{-1}$).
304 While affected by drought, Nit-S maintained a positive assimilation (Lowest
305 in Dec: $4.44 \pm 3.29 \mu\text{mol m}^{-2} \text{s}^{-1}$), demonstrating the highest resilience of the
306 three groups. Crucially, summer declines could neither be explained by
307 differences in leaf chlorophyll content nor leaf temperature (Supplementary
308 Figures S3 & S4), pointing to hydraulic constraints. Light sensitivity (k_m)
309 further differentiated these strategies. Nit-S retained the highest k_m
310 throughout spring (Sep: $353.14 \pm 120.29 \mu\text{mol m}^{-2} \text{s}^{-1}$, Figure 1) and early
311 summer (Jan: $185.9 \pm 45.15 \mu\text{mol m}^{-2} \text{s}^{-1}$), indicating a sustained capacity to
312 utilize high irradiance. Conversely, Rep-S k_m decreased continuously
313 throughout the same period (from 255.24 ± 52.62 down to 90.37 ± 113.87
314 $\mu\text{mol m}^{-2} \text{s}^{-1}$).

315 Adult resprouters (Nit-A) exhibited a conservative strategy, maintaining
316 stable k_m throughout the measurement period, and A_{sat} into December
317 (range: 8.53 – $12.34 \mu\text{mol m}^{-2} \text{s}^{-1}$), before eventually shifting into a lower
318 strategy in January ($2.7 \pm 1.28 \mu\text{mol m}^{-2} \text{s}^{-1}$).

319



320

321 *Figure 1: Saturating assimilation rate (A_{sat}) and light sensitivity or half-saturation light level*
 322 *(k_m) by plant type ('Nit-A': *P. nitida* adult; 'Nit-S': *P. nitida* seedling; 'Rep-S': *P. repens* seedling),*
 323 *measured throughout spring into summer. *P. nitida* is a resprouting and *P. repens* a reseeding*
 324 *species. Boxes indicate the interquartile range (IQR), middle bold black lines are medians of*
 325 *each of three leaf cohorts (Young, Mid, Old), which were pooled to illustrate the functional*
 326 *response of the whole plant. Whiskers represent 1.5-times the IQR. Only significant differences*
 327 *between months are shown. Full light response curves are provided in Supplementary Figure*
 328 *S2.*

329 **Differences in gas exchange and photochemical parameters:** To test
 330 Hypothesis 1 regarding carbon gain maximization, we analysed the binned
 331 categories of plant types by season (Figure 2). These results showed that
 332 Rep-S maintained significantly higher stomatal conductance (g_{sw} ,
 333 $0.11 \pm 0.043 \text{ mol m}^{-2} \text{s}^{-1}$) than *P. nitida* in all seasons (Nit-A: 0.056 ± 0.032 ; Nit-
 334 S: 0.070 ± 0.039) but summer (Rep-S: 0.010 ± 0.013 ; Nit-A: 0.043 ± 0.032 ; Nit-
 335 S: 0.041 ± 0.042), supporting the 'high risk-high reward' aspect of the
 336 reseeder strategy. Similarly, electron transport rate (ETR) and photosystem
 337 II efficiency (Φ_{PSII}) both dropped significantly below levels of *P. nitida* in
 338 summer ($78.9 \pm 56.5 \text{ } \mu\text{mol s}^{-1}$ vs. 166.0 ± 103.0 and 0.121 ± 0.097 vs.
 339 0.251 ± 0.171 for ETR & Φ_{PSII} in *P. repens* & *P. nitida*, respectively). Differences
 340 between Nit-A and Nit-S were not significant for Φ_{PSII} and, in most cases,
 341 ETR. However, Nit-S exhibited higher g_{sw} than Nit-A in all seasons (Nit-A:
 342 0.054 ± 0.037 ; Nit-S: 0.064 ± 0.041).

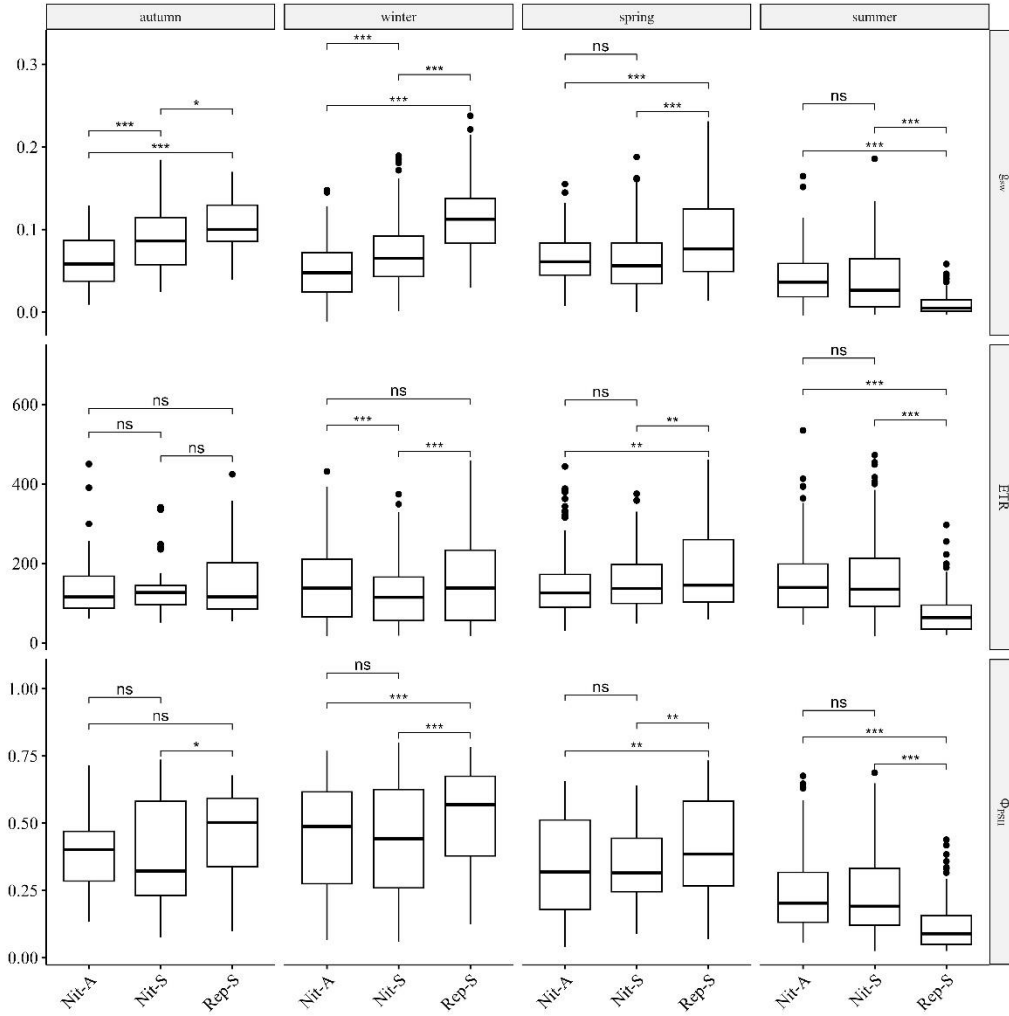


Figure 2: Differences in stomatal conductance (g_{sw} ; $\text{mol m}^{-2} \text{s}^{-1}$), electron transfer rate (ETR; $\mu\text{mol s}^{-1}$), quantum yield of photosystem II (Φ_{PSII} ; unitless) by season and type ('Nit-A': *P. nitida* adult; 'Nit-S': *P. nitida* seedling; 'Rep-S': *P. repens* seedling), where *P. nitida* is a resprouting and *P. repens* a reseeding species, comparing. Significance of differences is marked with asterisks ('ns': not significant) and was assessed using generalized linear mixed-effects models with post hoc pairwise contrasts (Holm-adjusted, Wald tests). Boxes indicate the interquartile range (IQR) with medians, whiskers represent 1.5-times the IQR, and outliers are indicated as points. Data were aggregated by season to reflect long-term trends and are defined as Autumn (Mar-May), Winter (Jun-Aug), Spring (Sep-Nov), and Summer (Dec-Feb).

Stomatal conductance model: As g_{sw} is understood to govern photosynthesis, a generalised additive model (GAM; deviance explained: 75.1%) was developed to further identify the key environmental factors governing g_{sw} using the median data of the 5 sampled individuals in each measurement (date, daytime, leaf age, plant group). The partial effects of leaf vapour pressure deficit (VPD_L), soil water content (SWC), seasonally

359 fluctuating CO₂ mole fraction in the air (x_{CO_2}) and photosynthetically active
360 radiation at the leaf (PAR_L) on g_{sw} are shown in Figure 3.

361 In support of Hypothesis 2 regarding physiological resilience and hydraulic
362 decoupling, the GAM results showed strong partial effects of VPD_L and PAR_L
363 on g_{sw} ($P < 0.01$; Figure 3a & b). However, VPD_L had a stronger effect in Rep-
364 S than in both *P. nitida* types. Interestingly, g_{sw} was insensitive to SWC
365 (Figure 3c & d; $P = 0.96$ & 0.50 for Nit-A & Nit-S, respectively) and x_{CO_2} in *P.*
366 *nitida* ($P = 0.68$ & 0.16 for Nit-A & Nit-S, respectively), while *P. repens*
367 responded strongly positively to increased SWC and x_{CO_2} ($P < 0.01$), whereby
368 the partial response of g_{sw} increased from $0.1 \text{ mol m}^{-2} \text{ s}^{-1}$ for x_{CO_2} starting at
369 395 ppm, plateauing at $0.2 \text{ mol m}^{-2} \text{ s}^{-1}$ for *ca.* 405 ppm. In essence, *P. nitida*
370 responds most strongly to heat and drought, while *P. repens* responds to
371 environmental parameters that include soil parameters. Note that these
372 responses were highly significant even though measurements of different
373 leaf ages and times of day were pooled, showing that results are robust
374 despite large variability.

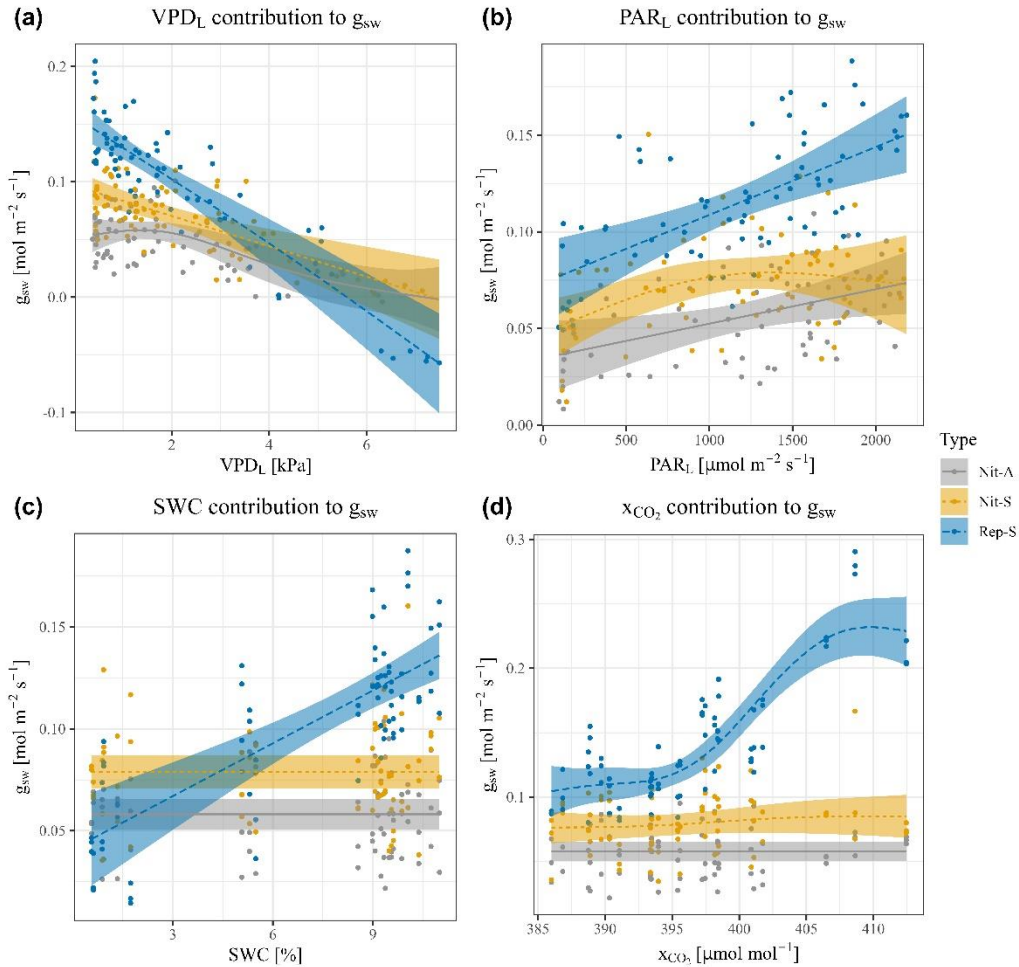


Figure 3: Partial effect plots of the Generalised Additive Model (GAM) of leaf vapour pressure deficit (VPD_L; panel a), photosynthetically active radiation at the leaf (PAR_L; panel b), soil water content (SWC; panel c), and CO₂ mole fraction in the air (x_{CO_2} ; panel d) on stomatal conductance (g_{sw}) for *P. nitida* (resprouter) adults (solid grey), *P. nitida* seedlings (dashed blue) and *P. repens* seedlings (reseeders; dashed orange), with the shaded area indicating 1 standard error bounds around the mean response. Each subplot describes the component effect of each parameter on g_{sw} if all other parameters are kept constant at their median value.

Discussion

While the relative prevalence of resprouting and reseeding strategies in fire-prone ecosystems has been well documented (Pausas et al. 2016; Altwegg et al. 2015; Clarke et al. 2013), the leaf-level physiological traits that underpin these divergent strategies in post-fire recovery have received comparatively little attention. However, these traits ultimately determine species resilience to fire, drought and heat under a changing climate, which is a particular challenge in many Mediterranean-type ecosystems (Keeley et

al. 2011). Our results provide a rare empirical comparison of the contrasting leaf-level strategies of three plant types: a reseeders (*Protea repens*) and resprouter (*Protea nitida*) seedling and a resprouter adult, ca. 2-3 years post-fire in a fire-prone summer drought environment. By evaluating closely related taxa, we ensured that the observed physiological differences reflect the regeneration strategies themselves rather than significant evolutionary divergence.

Stomatal regulation and photosynthetic divergence: Differences in CO₂ assimilation (A_{net}) between reseeders and resprouting species were predominantly driven by stomatal conductance (g_{sw}) rather than restricted by photochemical capacity, particularly during the more humid and cool seasons (autumn, winter & spring; Figs. 1 & 2). Minor differences in photochemical efficiency (Φ_{PSII} and ETR) and chlorophyll content between plant types (Fig. S3B) suggest that the photochemical machinery remained robust, leaving stomatal regulation as the primary driver of carbon gain (Fang et al. 2022; Keeley et al. 2011; Farquhar et al. 1982), similar to *Eucalyptus* seedlings where rapid gas exchange facilitates post-fire dominance (Rogers et al. 2024). Indeed, the reseeders (*P. repens*) exhibited significantly higher A_{net} and g_{sw} than resprouters during the cooler wetter seasons (autumn, winter, spring). Additionally, the reseeders reaching peak A_{sat} approximately a month earlier than the resprouter. However, this functional separation is most distinct during summer drought. In resprouters, the maintenance of high k_m (light sensitivity) and a relatively high g_{sw} with only gradual decrease in spite of the drought ‘shock’ of early summer suggests a sustained ‘sun-leaf’ physiology. This is consistent with an anisohydric strategy (Roman et al. 2015; Manzoni et al. 2013) and is facilitated by hydraulic access to deeper water via an established or rapidly developing tap root system, thus providing hydraulic safety. In contrast, the reseeders adopted a conservative, isohydric-like strategy, closing stomata and downregulating photosynthetic machinery (k_m) to minimize water loss.

421 This challenges the expectation that reseeders generally prioritize rapid
422 growth at the cost of hydraulic safety (e.g., Vilagrosa et al. 2014; Pratt et al.
423 2007; Jacobsen et al. 2007). For reseeders, survival during the critical first
424 years outweighs the ‘high-reward’ of continuous carbon gain.

425 **Life-history trade-offs in post-fire re-establishment:** Our results
426 highlight a clear reproduction/resilience trade-off and underscore the
427 importance of considering life-history strategy when evaluating plant
428 responses to climatic variability (Lawes and Clarke 2011; Lawes et al. 2022).
429 The reseeders’ investment in leaf area and high gas exchange during wet
430 months (Figs. 1 & 2) facilitates the rapid growth necessary to reach maturity
431 and seed production before the next fire (Hernández et al. 2011; Altwegg et
432 al. 2015; Keeley et al. 2011). However, this “high risk-high reward” strategy
433 becomes a liability during the seasonal summer drought, when a reliance on
434 shallow roots leads to a rapid shift into net carbon loss (Figure 3). In
435 contrast, resprouter seedlings generally maintained higher activity than
436 reseed seedlings during this period, but they likely did so at the cost of
437 increased water loss owing to greater stomatal conductance (Figures. 1 &
438 2), mitigated by prioritizing deeper water access and persistence (West et
439 al. 2012). By sacrificing leaf area and canopy density to develop deep roots
440 early on (Pausas and Keeley 2014), they effectively decouple their leaf-level
441 physiology from surface soil moisture deficits. While this results in lower
442 peak productivity compared to reseeders, it grants resprouter seedlings a
443 phenotypic plasticity needed to maintain activity during summer drought, a
444 factor not yet considered in existing models (Altwegg et al. 2015). These
445 divergent strategies facilitate an ecosystem-scale transition from reseed-
446 dominated photosynthesis during the cooler wet season to resprouter-
447 driven productivity during summer drought, a nuance that should be
448 incorporated into predictive frameworks like DGVMs.

449 **Seed legacy effects:** While resprouter seedlings generally adopt a
450 conservative water-use strategy, they exhibited unexpectedly high A_{sat} in

451 their old leaves, the first ones to emerge after fire (Figure 1; Supplementary
452 Figure S2) despite lower g_{sw} , Φ_{PSII} , ETR (Supplementary Figure S5), and
453 similar chlorophyll content (Supplementary Figure S3). This discrepancy
454 likely reflects an ontogenetic carry-over effect where early investment in
455 deep rooting, facilitated by larger seed reserves allows resprouters to
456 maintain hydraulic function. These results are consistent with the findings
457 of Clemente et al. (2005) as well as studies on the long-term impacts of seed
458 provisioning on seedling performance (Pausas and Keeley 2014; Altwegg et
459 al. 2015): Enhanced seed quality in resprouters may compensate for their
460 lower seed production relative to reseeders by facilitating more robust early
461 establishment, including the development of deep root systems that provide
462 a competitive advantage under limited water availability.

463 **The interplay of drought and elevated CO₂:** The requirement of reseeders
464 for reliable rainfall to ensure successful establishment leads to better
465 recruitment in relatively moist years. These may become increasingly rare
466 under future climate change scenarios (Pausas et al. 2016; Pausas and
467 Keeley 2014) and therefore lead to reduced reseeded recruitment.
468 Interestingly, their sensitivity of g_{sw} in ambient variations in seasonal
469 atmospheric CO₂ concentrations (x_{CO2} range: 385–415 ppm; Figure 3)
470 appears to provide a counterbalancing mechanism by enhancing
471 photosynthetic efficiency and growth rates under a future high-CO₂
472 scenario. Reseeders are currently CO₂-limited during their most active
473 (winter) season and an increase could afford them a competitive edge under
474 rising CO₂ despite the projected decline in favorable moisture conditions.
475 Importantly, the absence of a CO₂ response in resprouters (despite exposure
476 to the same seasonal fluctuations) suggests that the observed stimulation in
477 reseeders is not merely an artifact of seasonal fluctuations in CO₂
478 concentrations but rather reflects a genuine physiological response that
479 may influence post-fire community dynamics. In turn, the strong response
480 to CO₂ in reseeders may positively impact their water use efficiency. This

481 may also affect plant community composition under a changing climate, as
482 reseeders may benefit more from increasing atmospheric CO₂
483 concentrations, though only if fire return intervals do not increase
484 simultaneously.

485 **Conclusions:** Recent shifts in fire return intervals have important
486 implications for plant regeneration strategies. Because reseeding species
487 require longer favourable periods to establish and attain reproductive
488 maturity (Altwegg et al., 2015; Pausas & Keeley, 2014), resprouters are
489 predicted to gain a competitive edge under more frequent fires, as their
490 rapid recovery is supported by early investments in deep root systems and
491 persistence organs (Clarke et al. 2013). Moreover, fires occurring outside
492 the historical season tend to suppress post-fire recruitment in obligate
493 reseeders while favouring resprouters (Tangney et al. 2022). However, our
494 findings show that the reseeders' sensitivity to elevated CO₂ may provide a
495 mechanism to accelerate biomass accumulation during the cooler wet
496 season. This rapid early growth after fire (Rogers et al. 2024) could partially
497 offset their drought vulnerability under future warming scenarios, provided
498 it allows seedlings to develop sufficient root depth before the onset of
499 increasing fire return frequency or severe summer droughts.

500 The balance between these two variables is an important determinant for
501 improving predictions of vegetation shifts in fire-disturbed landscapes, e.g.
502 in DGVMs, as it can significantly drive plant community composition. Hence,
503 future research must disentangle these two drivers by incorporating more
504 fine-grained diurnal and seasonal measurements across multiple species,
505 years, and fire events. This broader temporal perspective would provide a
506 more comprehensive understanding of the dynamic physiological traits
507 throughout the life history of reseeders and resprouters. Such an approach
508 would further improve our findings of the sensitivity to CO₂ concentrations,
509 a crucial factor for predicting how these species might respond to a changing
510 climate.

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521 Author contributions

522 The study was conceived by JM & GM; ER and RC carried out ground
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525 Code and data availability

526 The data that support the findings is available from the corresponding
527 author upon reasonable request. Data analysis scripts are available at
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529 Competing Interests

530 None declared.

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